

ASSIGNMENT-2

E-commerce platform: Microservices & Event-Driven Architecture:

Microservices Architecture

The E-commerce platform is decomposed into domain specific, independent microservices:

- Product Service :- Manages product catalog & inventory
- Order Services: Handles order creation, updates & cancellations.
- Payment Services: Process secure customer payments
- Shipping Services: oversees delivery & logistics.
- Notification Service: Sends updates on orders, promotions etc.
- Customer Service: Manages profiles, authentication & preferences.

Event driven Architecture

- 1) Order placed → Emitted by order services when an Order is placed.
- 2) Payment processed → Emitted by payment services after processing payment.
- 3) Shipping started → triggered by shipping services after payment is confirmed
- 4) Notification → sent by Notification service based on relevant events.

Solid principles Applied:

- SRF: Each microservices has one responsibility
- OCP: New features (e.g. promotions) can be added without changing existing code.
- LSP: Services can be upgraded without breaking the system
- ISP: Services only specifying necessary APIs.
- DIP: High-level services rely on abstractions like events.

DRY & KISS Principle

• DRY:-

- use shared frameworks for event processing
- share common libraries (e.g. validation, auth) across services.

• KISS:-

- Keep services focused & simple
- Design clear, self-contained events.
- Minimised inter-service dependencies.